Data Structures

CSL202

Date: 16.09.2025 Quiz Set-A

1. **Write a function for Longest Valid Parentheses**

   Given a string containing just the characters '(' and ')', return the length of the longest valid (well-formed) parentheses

   Example 1: Input: s = "(()" Output: 2

   Explanation: The longest valid parentheses substring is "()".

   Example 2: Input: s = ")()())" Output: 4

   Explanation: The longest valid parentheses substring is "()()".

2. **Swap Left and Right Children of Every Node (Mirror Tree)**
   Write a program swapTree() that takes a binary tree and swaps the left and right children of every node.

Sample Input:	Sample Output:
1	1
/ \	/ \
2 3	3 2
/ \	/ \
4 5	5 4

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3. **Consider the following deque of characters where DEQUE is a circular array which is allocated six memory cells:**

   LEFT = 2,

   RIGHT = 4

   DEQUE: A, C, D

   Describe the deque while the following operations take place.

   (a) F is added to the right of the deque.

   (b). Two letters on the right are deleted.

   (c) K, L and M are added to the left of the deque.

   (d) One letter on the left is deleted.

   (e) R is added to the left of the deque.

   (f) S is added to the right of the deque.

   (g) T is added to the right of the deque.

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Data Structures

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Date: 16.09.2025 Quiz Set-B

1. **Given a Binary Search Tree (BST) and a key K, write a procedure to divide the BST into two trees:**

   a. One tree (T1) containing all keys < K

Data Structures

CSL202

Date: 16.09.2025 Quiz Set-B

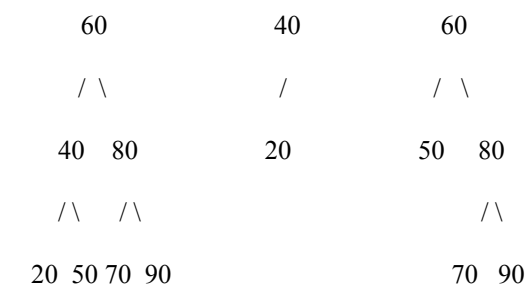
1. **Given a Binary Search Tree (BST) and a key K, write a procedure to divide the BST into two trees:**

   c. One tree (T1) containing all keys < K

b. Another tree (T2) containing all keys ≥ K

The procedure must preserve the BST property in both resulting trees.

Sample Input: K = 50 **Sample Output:**



2. **Consider the following deque where DEQUE is allocated 6 memory cells:**

LEFT = 2, **RIGHT** = 5

DEQUE: [London. Berlin, Rome, Paris]

Describe the deque. including LEFT and RIGHT, as the following operations take place:

(a) Athens is added on the left.

(b) Two cities are deleted from the right.

(c) Madrid is added on the left.

(d) Moscow is added on the right.

(e) Two cities are deleted from the right.

(f) A city is deleted from the left.

(g) Oslo is added on the left.

3. **Write a function to find Minimum Remove to Make Valid Parentheses**

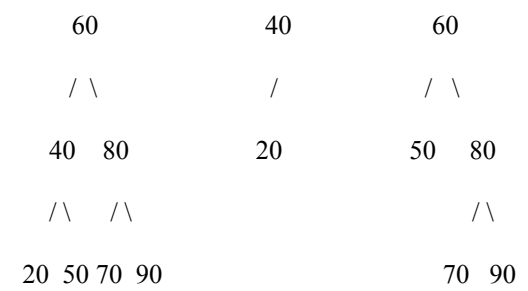
Given a string s of '(' , ')' and lowercase English characters. Your task is to remove the minimum number of parentheses ('(' or ')', in any positions) so that the resulting parentheses string is valid and return any valid string.

Formally, a parentheses string is valid if and only if:

d. Another tree (T2) containing all keys ≥ K

The procedure must preserve the BST property in both resulting trees.

Sample Input: K = 50 **Sample Output:**



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Formally, a parentheses string is valid if and only if:

a. It is the empty string, contains only lowercase characters, or

b. It can be written as AB (A concatenated with B), where A and B are valid strings, or

c. It can be written as (A), where A is a valid string.

Example :

Input: s = "lee(t(c)o)de)"

Output: "lee(t(c)o)de"

Explanation: "lee(t(co)de)" , "lee(t(c)ode)" would also be accepted.

Data Structures

CSL202

Date: 16.09.2025 Quiz Set-C

1. **Count Leaf Nodes in a Binary Tree**

   Write a program that counts the number of leaf nodes in a binary tree. A leaf node is a node that has no left or right child.

   Sample Input:

   ```
      10
     / \
    5   15
   / \
  2   7
   ```

   Sample Output:

   Number of leaf nodes = 3

2. **Given two Binary Search Trees (BSTs), write a procedure to merge them into a**

a. It is the empty string, contains only lowercase characters, or

b. It can be written as AB (A concatenated with B), where A and B are valid strings, or

c. It can be written as (A), where A is a valid string.

Example :

Input: s = "lee(t(c)o)de)"

Output: "lee(t(c)o)de"

Explanation: "lee(t(co)de)" , "lee(t(c)ode)" would also be accepted.

Data Structures

CSL202

Date: 16.09.2025 Quiz Set-C

1. **Count Leaf Nodes in a Binary Tree**

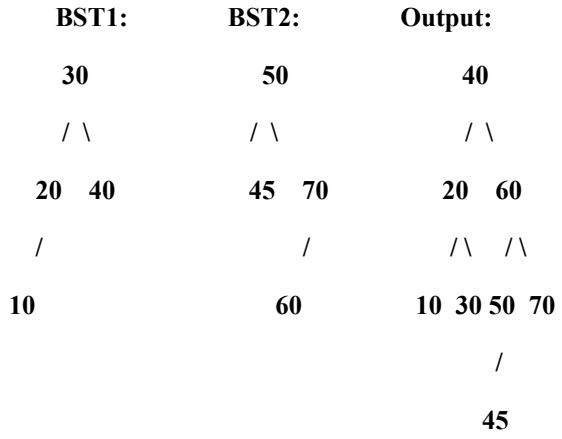
   Write a program that counts the number of leaf nodes in a binary tree. A leaf node is a node that has no left or right child.

   Sample Input:

   ```
      10
     / \
    5   15
   / \
  2   7
   ```

   Sample Output:

single balanced BST without inserting nodes one at a time from one tree into another. The procedure should take advantage of the fact that the inorder traversal of a BST gives a sorted sequence.



3. **A deque is implemented using a circular array of 6 memory cells.**

   Initially:

   LEFT = 0 RIGHT = 2

   DEQUE = [A, B, C]

Describe the deque. including LEFT and RIGHT, as the following operations take place:

(a) Add D to the right.

(b) Add E to the left.

(c) Delete two elements from the left.

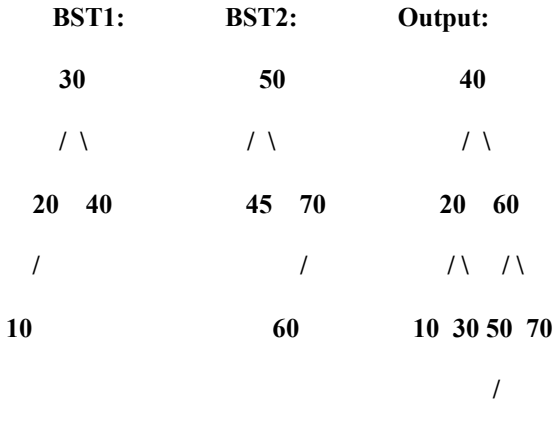
(d) Add F and G to the right.

(e) Delete three elements from the right.

(f) Add H to the left.

Number of leaf nodes = 3

2. **Given two Binary Search Trees (BSTs), write a procedure to merge them into a single balanced BST without inserting nodes one at a time from one tree into another. The procedure should take advantage of the fact that the inorder traversal of a BST gives a sorted sequence.**



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